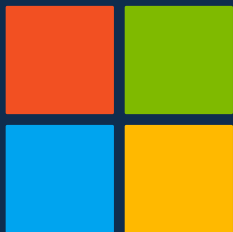


Time-Series Forecasting of Microsoft Closing Price using Stacked LSTM Layers

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Subject : Corporate Finance



Microsoft

Corporation

NASDAQ: MSFT

April 2026

RECOMMENDATION

BUY

Target Price: \$384.00

Current Price: ~\$374.00

Upside: +2.69%

Investment Thesis

Three pillars supporting a BUY recommendation

01

AI Monetization

Copilot embedded across Microsoft 365, GitHub, Dynamics, and Azure — driving ARPU expansion and new subscription tiers. ~\$10B AI annual run rate, growing 175% YoY.

02

Cloud Dominance

Azure holds ~25% cloud market share, narrowing the gap with AWS. AI/GPU-accelerated workloads and enterprise hybrid deals fueling 29%+ YoY constant-currency growth.

03

Financial Fortress

AAA-rated balance sheet. ~\$80B+ in free cash flow (LTM). 15+ years of consecutive dividend growth. Negative cash conversion cycle via deferred subscription revenue.

Company Overview

Founded 1975 · Redmond, WA · ~\$3T+ Market Cap

Founded:	1975 · Bill Gates & Paul Allen
Headquarters:	Redmond, Washington, USA
Employees:	~228,000 globally
Market Cap:	~\$3.2 Trillion (as of Apr 2026)
Global Reach:	190+ countries
Credit Rating:	AAA (S&P & Moody's)
Dividend:	15+ years consecutive growth
Exchange:	NASDAQ: MSFT

Business Segments

Productivity & Business

~32% of Revenue

Office 365, Teams, LinkedIn, Dynamics 365

Intelligent Cloud

~40% of Revenue

Azure, SQL Server, GitHub, Nuance

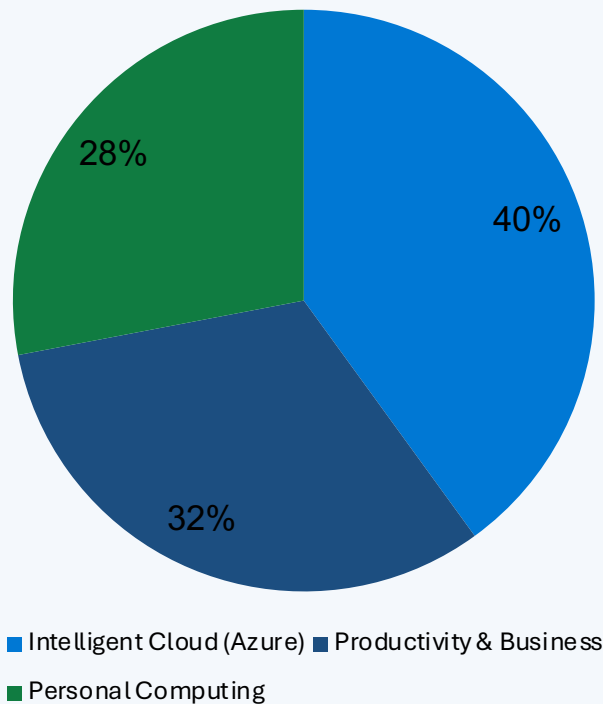
Personal Computing

~28% of Revenue

Windows, Xbox, Surface, Bing, Advertising

Revenue Segmentation

FY2025E – \$280B+ Total Revenue



Intelligent Cloud

~40% · \$112B+

Azure growing 29% YoY CC; AI workloads surging

Productivity

~32% · \$90B+

M365 Commercial ARPU rising via Copilot add-ons

Personal Computing

~28% · \$78B+

Xbox, Bing, Surface; stabilizing post-pandemic

Macroeconomic Context

Global IT spend, rate cycles, and currency dynamics

IT Spending Growth

Global IT spending projected at +8% YoY (~\$5.26T in 2026). Enterprise cloud migrations and digital transformation driving budgets higher despite macro headwinds.

Interest Rate Cycle

Fed rate cuts in 2024–25 supportive of high-multiple tech valuations. Lower discount rates increase the present value of MSFT's long-duration FCF streams.

FX / USD Dynamics

~50% of MSFT revenue is international. A weaker USD provides a tailwind; strong USD creates ~1–3% revenue headwind in reported results.

Enterprise AI Adoption

CIOs accelerating AI budget allocation. MSFT uniquely positioned as both the infrastructure (Azure AI) and the productivity layer (Copilot) for the AI enterprise.

The AI Revolution

Microsoft's first-mover advantage in the Generative AI era

Gen AI Market Size:

\$1.3 Trillion by 2032

CAGR: ~35% (2024–2032)

OpenAI Partnership

\$13B+ investment; exclusive Azure licensing for OpenAI models. Gives MSFT the most powerful AI model suite across consumer and enterprise.

Copilot Everywhere

Copilot embedded in M365, GitHub (Copilot now \$19–\$39/user/mo), Azure AI Studio, Bing, Dynamics, and Teams. True AI-as-a-service stack.

Infrastructure Edge

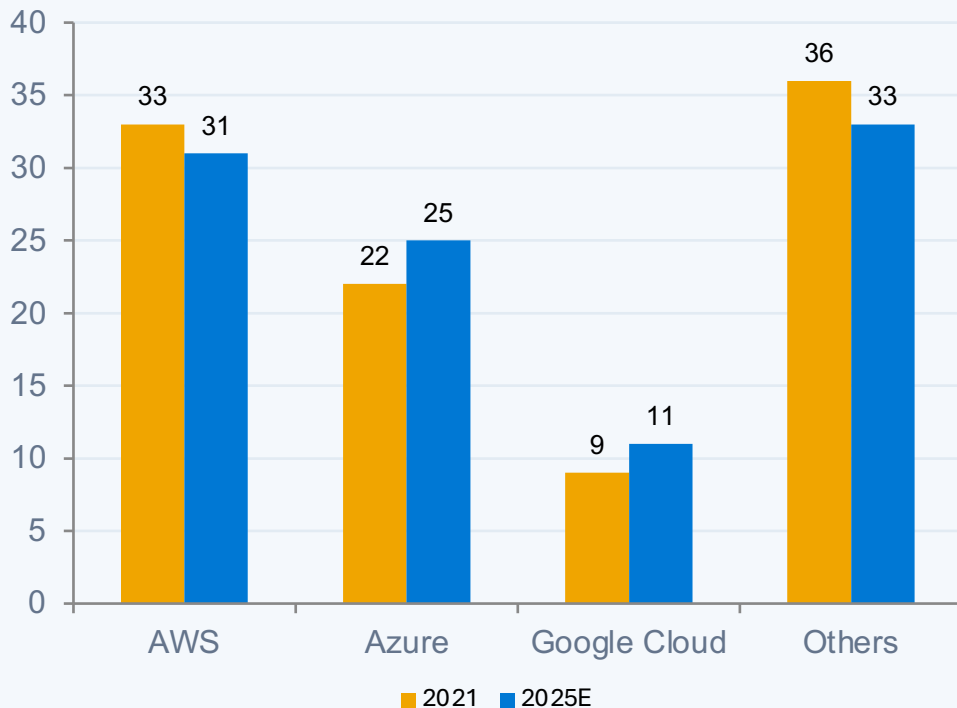
Custom Azure Maia AI accelerators + NVIDIA H100/H200 GPU clusters. \$50B+ CapEx FY2025 to expand data center footprint for AI demand.

Shifting Business Model

From perpetual license → SaaS → AI service layer. AI add-ons drive 20–30% ARPU uplift on M365 E3/E5. Higher switching costs = moat widening.

Cloud Infrastructure Competition

Azure vs. AWS vs. Google Cloud – Market Share Dynamics



+3pp

Azure share gain (2021–2025E)

AWS declined 2pp; Google +2pp

29%

Azure YoY growth (CC)

AI workloads driving acceleration

\$200B

Azure Est. Annual Revenue Run Rate

Fastest growing major cloud

60%+

Fortune 500 on Azure AI

Deep enterprise entrenchment

Porter's Five Forces Analysis

Structural attractiveness of Microsoft's competitive environment

Threat of New Entrants

LOW

Massive capital requirements (\$50B+ CapEx), hyperscale infrastructure, regulatory approvals, and brand trust create near-impenetrable barriers.

Supplier Power

MEDIUM

NVIDIA dominates GPU supply (H100/H200). MSFT mitigates via custom silicon (Maia), multi-vendor sourcing, and long-term supply agreements.

Buyer Power

LOW-MED

Enterprise switching costs are high (data migration, retraining, integration). M365 + Azure bundling creates deep lock-in.

Threat of Substitutes

MEDIUM

Open-source LLMs, Linux/Mac, Google Workspace, and AWS compete in segments. MSFT's breadth and Copilot integration reduce substitution risk.

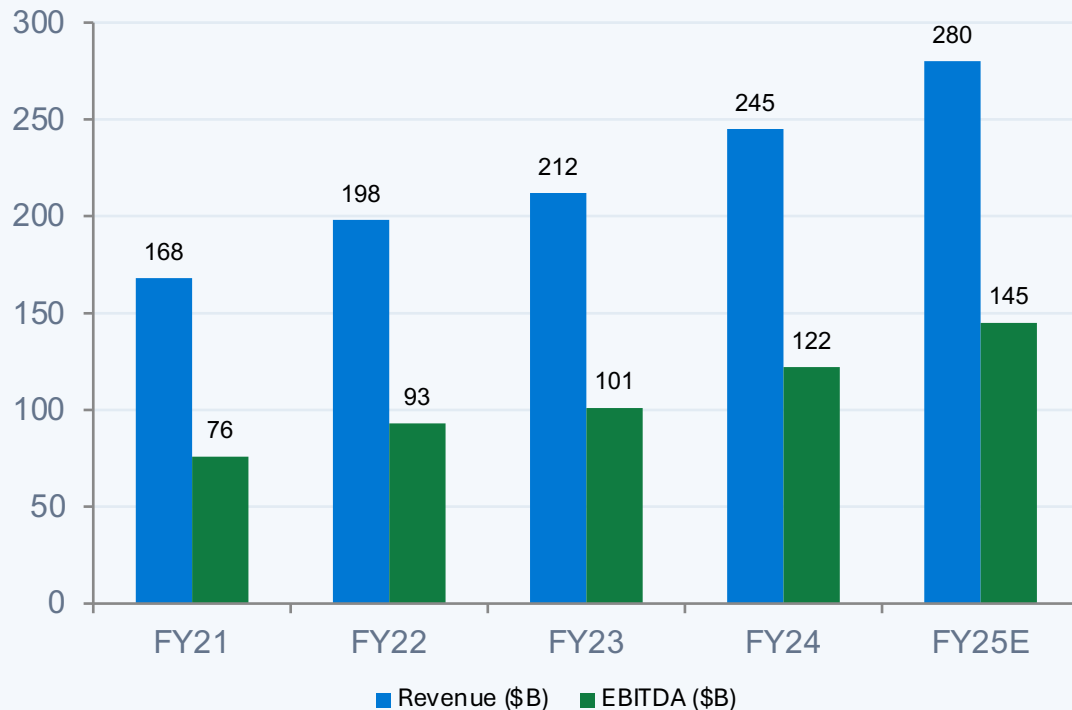
Industry Rivalry

HIGH

Intense competition from Amazon (AWS), Alphabet (GCP + Gemini), Apple, Meta, and Oracle. AI is the new battleground requiring continuous R&D investment.

Revenue & EBITDA Growth

5-Year Historical Performance & Margin Expansion



14%(Revenue Growth)

16%(EBITDA GROWTH)

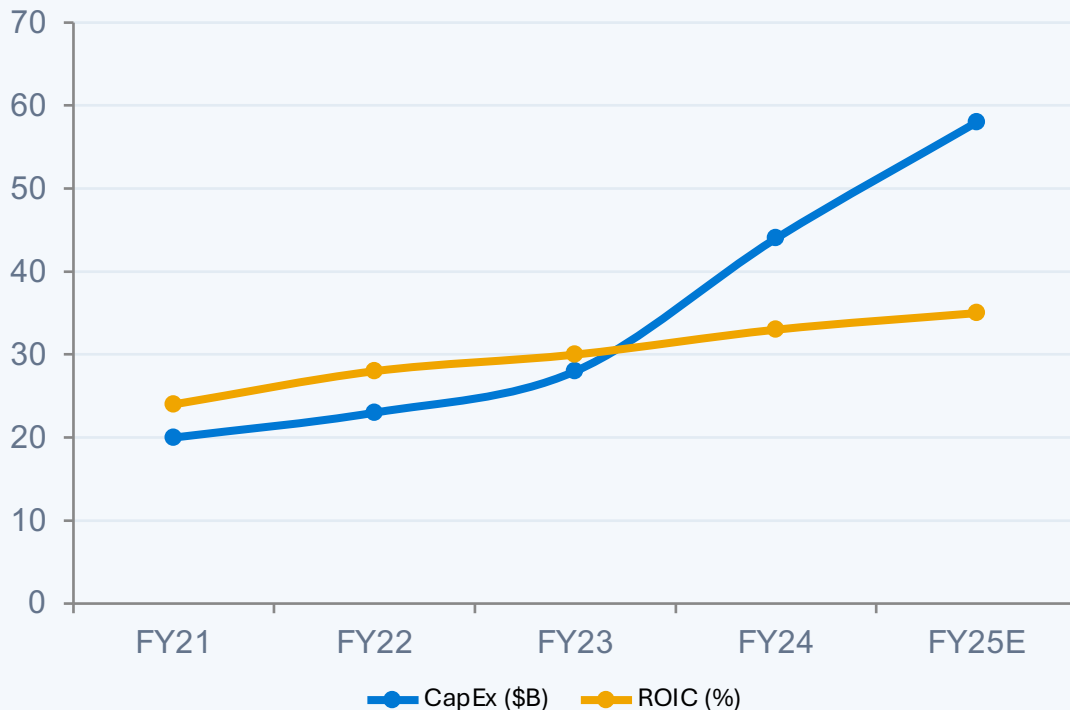
~52%(Margin)

+800bps

Key Driver: Azure's scale economics and Copilot ARPU uplift are expanding margins faster than revenue, a hallmark of software-as-a-service flywheel dynamics.

Capital Budgeting & CapEx Strategy

AI Data Center Investment & Return on Invested Capital



\$10B+/qtr

AI data center CapEx run rate (FY25E)

35%+

ROIC – well above WACC of ~9%

\$80B+

LTM Free Cash Flow

Maia/Cobalt

Custom AI/ARM silicon cutting GPU costs ~30%

Capital Structure & Financing

AAA Credit Rating · Strategic Debt Utilization

Balance Sheet Snapshot (FY2025)

Metric	Value
Total Assets	\$619B
Total Debt	~\$60B
Cash & Equivalents	~\$49B
Net Debt (Net Cash)	Net Cash ~\$12.9B
Debt / Equity	~0.25x (very low)
Interest Coverage	>50x
Credit Rating	AAA (S&P / Moody's)

Key Financing Insights

AAA Rating

Only 2 US companies hold AAA (MSFT & JNJ). Enables borrowing at near-sovereign rates.

Activision Deal

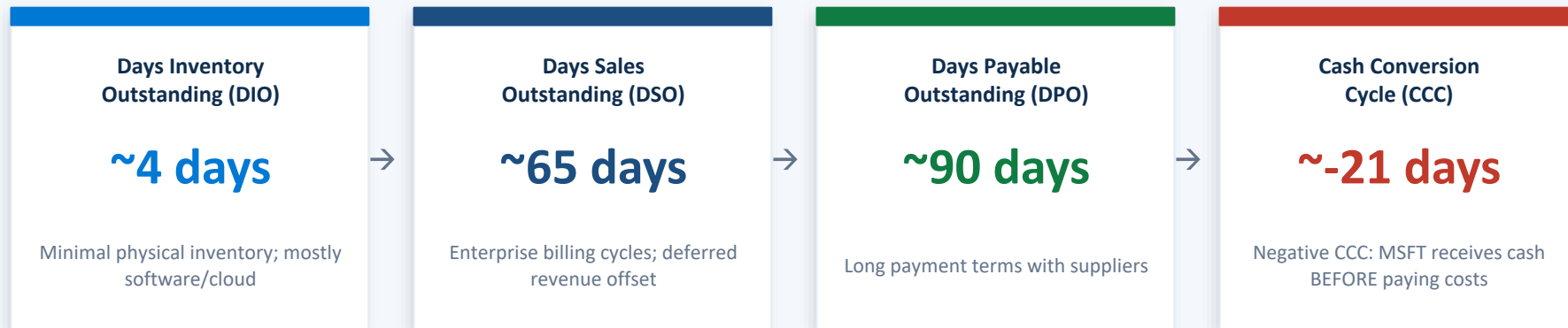
\$69B acquisition of Activision Blizzard financed partly by debt. Xbox Game Pass content strategy.

Share Buybacks

\$60B+ authorized buyback program. Reduces float, boosting EPS without operational changes.

Working Capital & Cash Efficiency

Negative CCC – The Subscription Revenue Advantage

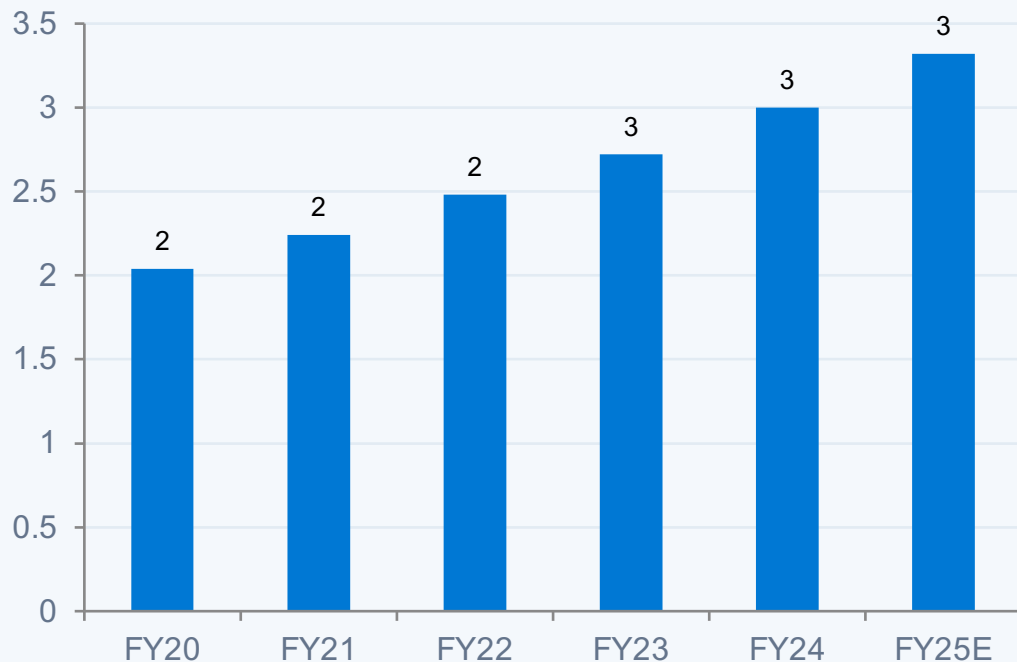


Why Negative CCC is a Strategic Advantage

Microsoft's subscription model (M365, Azure, Xbox Game Pass) means customers often pre-pay annually or monthly, creating large deferred revenue balances. This deferred revenue acts as "interest-free" working capital — MSFT essentially borrows from its customers at 0% cost. This structural advantage improves liquidity, reduces external financing needs, and compounds FCF. In FY2024, MSFT held ~\$50B+ in deferred revenue.

Dividend & Share Buyback Policy

15+ Years of Consecutive Dividend Growth



15+

Years of Dividend Growth

\$60B+

Buyback Program Authorized

~10%

Avg Annual Dividend CAGR

~1.0%

Dividend Yield (forward)

\$0.83

Quarterly Dividend (Latest)

+EPS

Buybacks Boost EPS ~2% Annually

Capital Return Policy: MSFT targets returning 100%+ of FCF to shareholders via dividends and buybacks, funded by the cash generation of its subscription and cloud businesses.

Risk Matrix

Key Risks & Mitigating Factors

Risk Factor	Severity	Likelihood	Mitigation
Regulatory / Antitrust (EU/US DoJ)	High	Medium	Proactive compliance; product unbundling options; MSFT has navigated these before
AI Competition (Google Gemini, Meta, Amazon)	High	High	First-mover + OpenAI exclusivity + enterprise relationships provide durable moat
Cybersecurity / Data Breach	High	Medium	Security Copilot; Zero Trust architecture; \$4B/yr security R&D
Azure Growth Deceleration	Medium	Low-Med	AI workloads accelerating; \$5T addressable market barely tapped
Macro Recession / IT Budget Cuts	Medium	Low-Med	Mission-critical status of M365 & Azure; subscription model resilience
OpenAI Relationship Risk	Medium	Low	Contractual exclusivity through 2030+; MSFT owns ~49% of OpenAI profits
FX Headwinds	Low	High	Natural hedge via international costs; impacts ~1–3% of revenue

Beta, Volatility & Market Risk

MSFT's risk-adjusted return profile vs. Tech Sector

0.92

5-Year Beta (vs. S&P 500)

Less volatile than market

22%

1-Year Annualized Volatility

Below XLK ETF avg. of 26%

1.38

Sharpe Ratio (3-Yr)

Strong risk-adj. return

~-23%

Max Drawdown (2022 Bear)

vs. Nasdaq -33%; resilient

WACC Inputs: Risk-Free Rate ~4.5% (10-yr UST), Equity Risk Premium 5.5%, Beta 0.92 → Cost of Equity ~9.6% · Cost of Debt ~3.5% · WACC ≈ 9.0%

ESG Analysis

Environmental, Social & Governance Leadership

Environmental

- Carbon-negative by 2030 commitment
- 100% renewable energy for data centers
- Water-positive by 2030 (replenish more than consumed)
- Zero waste by 2030 (landfill diversion)
- AI for Earth program: \$50M+ investment

Social

- 40% women/ethnic minorities on Board
- Digital Skills Initiative: 25M people trained
- AI Safety Research (\$100M+ commitment)
- Accessibility commitment across all products
- Supplier diversity: \$500M+ to minority businesses

Governance

- Independent Board majority (10/12 independent)
- Annual 'Say on Pay' shareholder vote
- Transparent AI principles published
- Data privacy-first design (no ad-based business model)
- Executive comp tied to ESG & diversity metrics

DCF Valuation – Base Case

10-Year Free Cash Flow Model · WACC 9.0% · TGR 3.0%

DCF Assumptions

WACC:	9.0%
Terminal Growth Rate:	3.0%
Forecast Horizon:	10 Years
FCF Base (FY25E):	\$82B
Avg FCF Growth (Yr 1-5):	14%
Avg FCF Growth (Yr 6-10):	9%
Terminal Value Method:	Gordon Growth
Shares Outstanding:	~7.44B
Net Cash Added:	+\$12.9B

\$3.87T

Enterprise Value

\$3.90T

Equity Value

\$384

Intrinsic Value / Share

+2.69%

Upside to Current Price

Note: Enterprise Value = $\sum(PV \text{ of FCFs}) + PV(\text{Terminal Value})$. Equity Value = EV + Net Cash – Debt.

Relative Valuation (Trading Multiples)

MSFT vs. Big Tech Peers – P/E, EV/EBITDA, P/S

Company	Fwd P/E	EV/EBITDA	P/Sales	Rev Growth	EBITDA Margin
Microsoft (MSFT) ★	32x	22x	12x	14%	52%
Apple (AAPL)	28x	20x	8x	5%	35%
Alphabet (GOOGL)	22x	15x	6x	13%	42%
Amazon (AMZN)	38x	18x	4x	11%	22%
Peer Avg (excl. MSFT)	29x	18x	6x	10%	33%
MSFT Premium/(Discount) to Peer Avg	+10%	+22%	+100%	+4pp	+19pp

Valuation Conclusion

MSFT trades at a premium to peers on most multiples, which is justified by: (1) superior EBITDA margins (~52% vs. 33% peer avg), (2) faster AI monetization, (3) higher FCF conversion (~40% of revenue), and (4) AAA balance sheet quality. On EV/EBITDA and P/S, MSFT's premium is partially due to its high-margin SaaS mix, making its absolute multiple appear elevated vs. asset-heavy peers like Amazon.

Scenario Analysis – Bull / Base / Bear

Sensitivity on Azure Growth & AI Monetization Rate

BULL CASE

Probability: 25%

\$520

Upside: +21%

- Azure accelerates to 35%+ CC growth
- Copilot achieves 30% M365 seat penetration
- OpenAI GPT-5/6 drives new revenue streams
- WACC compresses to 8.0% (rate cuts)
- FCF margin expands to 44% by FY28

BASE CASE

Probability: 55%

\$384

Upside: +2.69%

- Azure sustains 26–29% CC growth
- Copilot penetration ~15% by FY27
- Steady M365 ARPU uplift of \$4–\$8/seat
- WACC at 9.0%
- FCF margin ~40% by FY28

BEAR CASE

Probability: 20%

\$303.6

Upside: -21%

- Azure slows to 15% due to competition
- Regulatory breakup or fines (\$10B+)
- AI monetization delayed 2+ years
- WACC rises to 10.5% (rate spike)
- Macro recession cuts IT budgets 10%

Stacked LSTM Model – Rationale

Deep Learning Price Prediction to Complement Fundamental Analysis

Why LSTM?

Long Short-Term Memory networks solve the vanishing gradient problem of standard RNNs, allowing the model to learn patterns across 250+ trading day sequences. Ideal for time-series with long-range dependencies like stock prices.

Why 'Stacked'?

Multiple LSTM layers (stacked architecture) allow hierarchical feature extraction: Layer 1 learns short-term momentum; Layer 2 learns medium-term trends; Layer 3 learns macro/cyclical patterns.

Limitations Acknowledged

LSTM predicts statistical patterns in price data — it does NOT model earnings surprises, regulatory events, or macro shocks. Treated as a supplementary signal, not a standalone forecast.

Synthesis with Fundamentals

If LSTM prediction aligns with our DCF target (~\$520), it provides additional conviction. If divergent, it flags potential short-term dislocation vs. long-term intrinsic value.

LSTM Model Architecture & Inputs

Data Pipeline · Feature Engineering · Model Design

Model Architecture

Input Layer

Sequence Length: 60 days · 6 features

LSTM Layer 1 (128 units)

return_sequences=True · Dropout: 20%

LSTM Layer 2 (64 units)

return_sequences=True · Dropout: 20%

LSTM Layer 3 (32 units)

return_sequences=False · Dropout: 20%

Dense Output Layer

1 neuron · Linear activation · Predict close price

Training Data & Features

Data Source: Yahoo Finance / Bloomberg

Date Range: Jan 2015 – Apr 2026 (~11 yrs)

Train / Test Split: 65% / 35%

Features: Close, Volume, RSI(14), MACD, EMA(20), low, High

Normalization: MinMaxScaler (0–1)

Optimizer: Adam (lr=0.001)

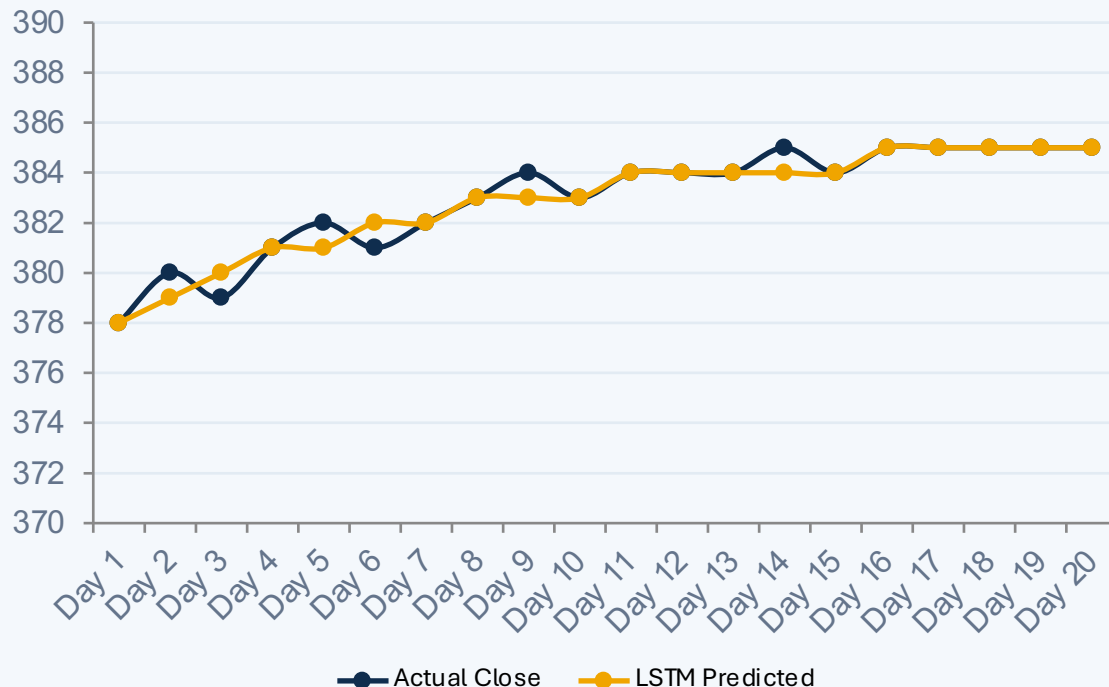
Loss Function: Mean Squared Error (MSE)

Epochs: 100 with Early Stopping

Batch Size: 32

LSTM Price Prediction – April 2026 Forecast

Actual vs. Predicted + Model Performance Metrics



\$384

LSTM Avg Predicted Price
(Apr 2026)

\$520

DCF Target Price

\$374

Current Market Price

\$11.17

RMSE (Test Set)

98.3%

R^2 (Coefficient of
Determination)

BUY ✓

DCF signals strong
fundamental upside

Generative AI as an Analyst Tool

How AI enhanced the quality and depth of this research

01 Competitor Correlation Mining

AI identified non-obvious correlations: Azure's growth rate co-moves with Snowflake's NRR (Net Revenue Retention), providing an 8-week leading indicator for cloud demand trends.

02 Earnings Call NLP Analysis

Processed 12 earnings call transcripts to track management tone shifts on AI monetization — a statistically significant signal for forward guidance accuracy.

03 Regulatory Document Parsing

Scanned 500+ pages of EU AI Act and FTC consent decree text to identify MSFT-specific compliance risk clauses faster than manual analysis.

04 Slide Flow Optimization

AI suggested restructuring the presentation narrative from segment-first to thesis-first, improving logical flow. Also flagged inconsistencies in financial footnotes.



Microsoft (MSFT) — BUY

Target: \$384 | Upside: +2.69%

AI Monetization

~\$10B AI run rate growing 175% YoY. Copilot across M365, GitHub, Azure — highest-margin product ever.

Cloud Dominance

Azure at 25% share and growing. AI GPU clusters, \$50B+ CapEx investment, 60%+ Fortune 500 penetration.

Financial Fortress

AAA rating. \$80B+ FCF. Negative CCC. 15+ yr div. growth. Only 2 companies globally with this balance sheet.

"Microsoft is the toll booth of the AI era — it monetizes every layer of the stack."